# EXPERIMENT – 6

**AIM:** To create a stored procedure `transfer_employee(emp_id, new_dept_id)` with validation and error handling, and to implement database triggers for salary validation and audit logging. The procedure and triggers are tested using normal and edge-case inputs.

## SOURCE CODE:

**USING EXPERIMENT – 3 DATABASE**

## SQL QUERIES:

**STORED PROCEDURE:** `transfer_employee`

The procedure will:

1. Check whether the employee exists.
2. Check whether the new department exists.
3. Check whether the employee is already in that department.
4. Transfer the employee.
5. Handle errors using `SQLEXCEPTION`.
6. Roll back the transaction if an error occurs.

```
DELIMITER //

CREATE PROCEDURE transfer_employee(
    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE v_employee_count INT DEFAULT 0;
    DECLARE v_department_count INT DEFAULT 0;
    DECLARE v_current_dept INT DEFAULT 0;

    -- Error handler
    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        ROLLBACK;
        SELECT 'Error occurred. Transaction rolled back.' AS
message;
    END;

    START TRANSACTION;
```

```
    -- Check whether employee exists
    SELECT COUNT(*)
    INTO v_employee_count
    FROM Employee
    WHERE emp_id = p_emp_id;

    IF v_employee_count = 0 THEN
        ROLLBACK;
        SELECT 'Error: Employee does not exist.' AS message;

    ELSE

        -- Check whether department exists
        SELECT COUNT(*)
        INTO v_department_count
        FROM Department
        WHERE dept_id = p_new_dept_id;

        IF v_department_count = 0 THEN
            ROLLBACK;
            SELECT 'Error: Department does not exist.' AS
message;

        ELSE

            -- Get current department
            SELECT dept_id
            INTO v_current_dept
            FROM Employee
            WHERE emp_id = p_emp_id;

            -- Check if already in requested department
            IF v_current_dept = p_new_dept_id THEN
                ROLLBACK;
                SELECT 'Error: Employee is already in this
department.'
                AS message;

            ELSE

                -- Transfer employee
                UPDATE Employee
                SET dept_id = p_new_dept_id
                WHERE emp_id = p_emp_id;

                COMMIT;

                SELECT
                    'Employee transferred successfully.'
                    AS message;
```

```
            END IF;

        END IF;

    END IF;

END //

DELIMITER ;
```

**TESTING THE STORED PROCEDURE**

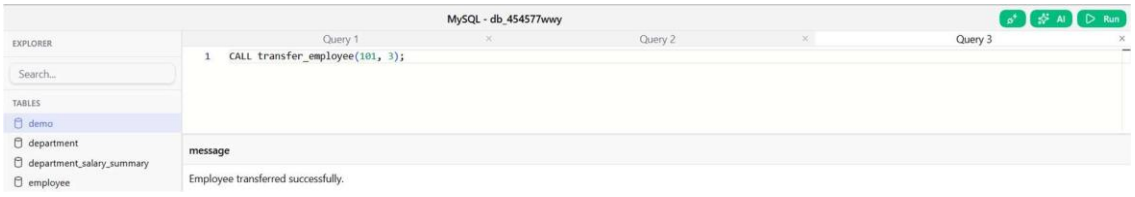
```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
WHERE e.emp_id = 101;
```

**OUTPUT**:




```
CALL transfer_employee(101, 3);
```

**OUTPUT**:



```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
WHERE e.emp_id = 101;
```

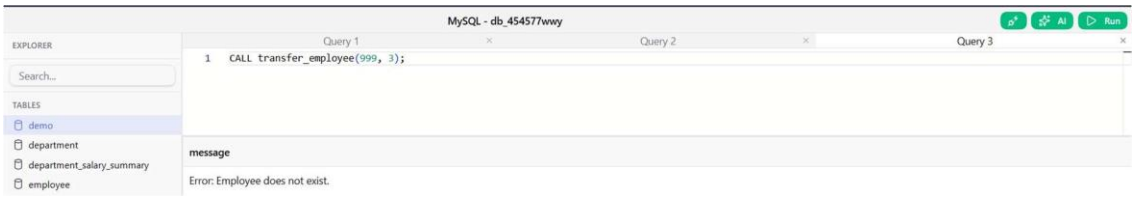
**OUTPUT**:




**EDGE CASE – EMPLOYEE DOES NOT EXIST**

```
CALL transfer_employee(999, 3);
```

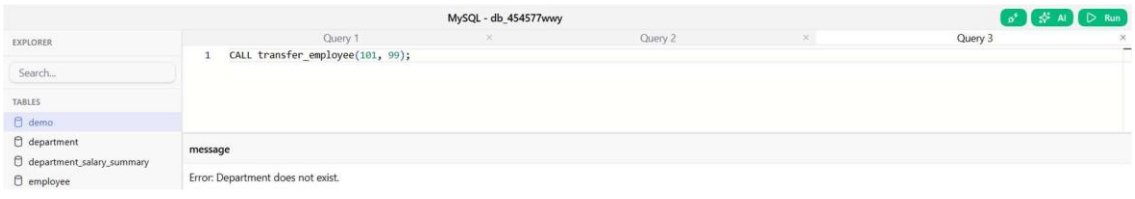
**OUTPUT**:




**EDGE CASE – DEPARTMENT DOES NOT EXIST**

```
CALL transfer_employee(101, 99);
```

**OUTPUT**:



**EDGE CASE – SAME DEPARTMENT**

```
CALL transfer_employee(101, 3);
```

**OUTPUT**:




**SALARY VALIDATION TRIGGER**

Let, salary must be >= 15000

```
DELIMITER //

CREATE TRIGGER validate_employee_salary
BEFORE INSERT ON Employee
FOR EACH ROW
BEGIN

    IF NEW.salary < 15000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be at least 15000';
    END IF;

END //

DELIMITER ;
```

**TEST SALARY VALIDATION**

```
INSERT INTO Employee
(emp_id, emp_name, gender, salary, hire_date, dept_id)
VALUES
(131, 'Test Employee', 'M', 10000, '2026-09-25', 1);
```

**OUTPUT**:

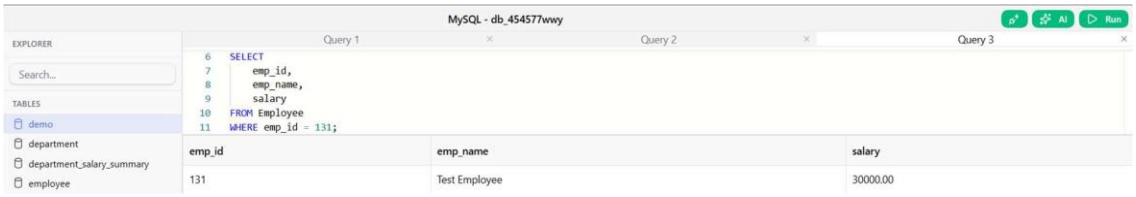


**TEST VALID SALARY**

```
INSERT INTO Employee
(emp_id, emp_name, gender, salary, hire_date, dept_id)
VALUES
(131, 'Test Employee', 'M', 30000, '2026-09-25', 1);

SELECT
    emp_id,
    emp_name,
    salary
FROM  Employee
WHERE emp_id = 131;
```

**OUTPUT**:




**SALARY VALIDATION FOR UPDATE**

```
DELIMITER //

CREATE TRIGGER validate_employee_salary_update
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN

    IF NEW.salary < 15000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be at least 15000';
    END IF;

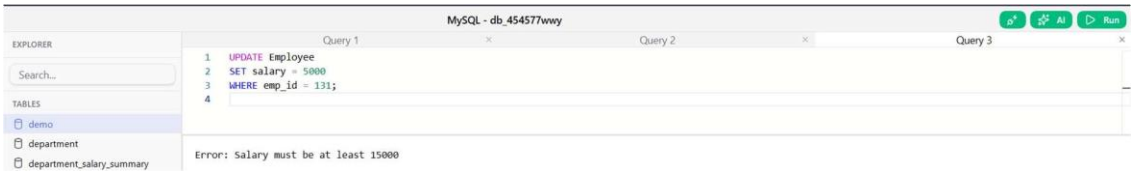
END //

DELIMITER ;
```

**TEST INVALID SALARY UPDATE**

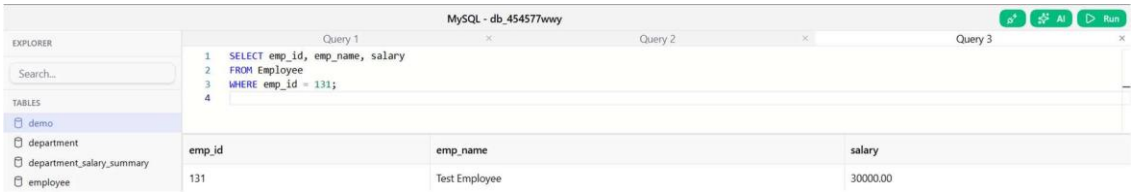
```
UPDATE Employee
SET salary = 5000
WHERE emp_id = 131;
```

**OUTPUT**:




```
SELECT emp_id, emp_name, salary
FROM Employee
WHERE emp_id = 131;
```

**OUTPUT**:




**CREATE AUDIT TABLE**

```
CREATE TABLE Salary_Audit (
    audit_id INT AUTO_INCREMENT PRIMARY KEY,
    emp_id INT,
    old_salary DECIMAL(10,2),
    new_salary DECIMAL(10,2),
    changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    action VARCHAR(30)
);
```

**AUDIT LOGGING TRIGGER**

```
DELIMITER //

CREATE TRIGGER salary_audit_trigger
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
```

```
    IF OLD.salary <> NEW.salary THEN

        INSERT INTO Salary_Audit
        (
            emp_id,
            old_salary,
            new_salary,
            action
        )
        VALUES
        (
            NEW.emp_id,
            OLD.salary,
            NEW.salary,
            'SALARY UPDATED'
        );

    END IF;

END //

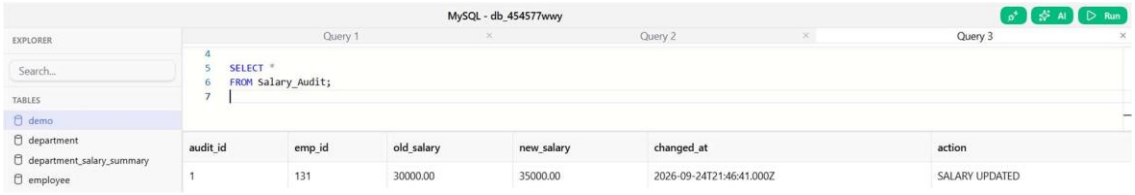
DELIMITER ;
```

**TEST AUDIT TRIGGER**

```
UPDATE Employee
SET salary = 35000
WHERE emp_id = 131;

SELECT *
FROM Salary_Audit;
```

**OUTPUT**:



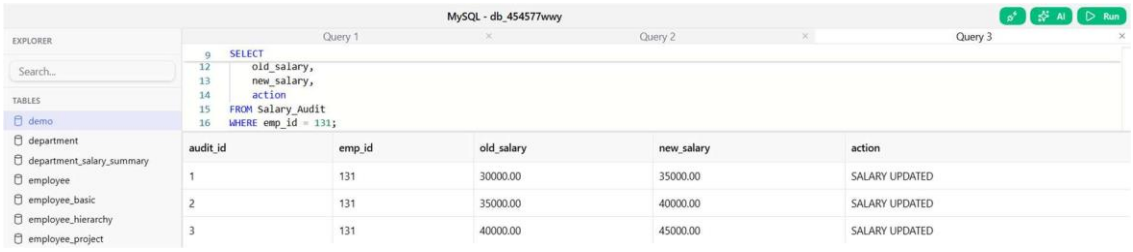
**TEST MULTIPLE SALARY CHANGES**

```
UPDATE Employee
SET salary = 40000
WHERE emp_id = 131;

UPDATE Employee
SET salary = 45000
WHERE emp_id = 131;

SELECT
    audit_id,
    emp_id,
    old_salary,
    new_salary,
    action
FROM Salary_Audit
WHERE emp_id = 131;
```

**OUTPUT**:




**TEST EDGE CASE – SALARY UNCHANGED**

```
UPDATE Employee
SET salary = 45000
WHERE emp_id = 131;
```

No new audit record is created

**TEST EDGE CASE – NULL SALARY**

```
DROP TRIGGER validate_employee_salary;

DELIMITER //

CREATE TRIGGER validate_employee_salary
BEFORE INSERT ON Employee
FOR EACH ROW
BEGIN
```

```
    IF NEW.salary IS NULL OR NEW.salary < 15000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Salary cannot be NULL and must be at least 15000';
    END IF;

END //

DELIMITER ;

INSERT INTO Employee
(emp_id, emp_name, gender, salary, hire_date, dept_id)
VALUES
(132, 'Null Salary', 'F', NULL, '2026-09-25', 1);
```

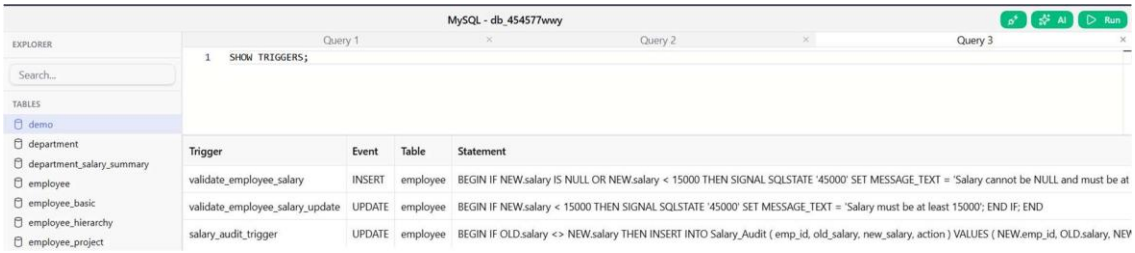
**OUTPUT**:




**CHECK ALL TRIGGERS**

```
SHOW TRIGGERS;
```

**OUTPUT**:



**CHECK STORED PROCEDURE**

```
SHOW PROCEDURE STATUS
WHERE Db = 'db_454577wwy';
```

**OUTPUT**:

